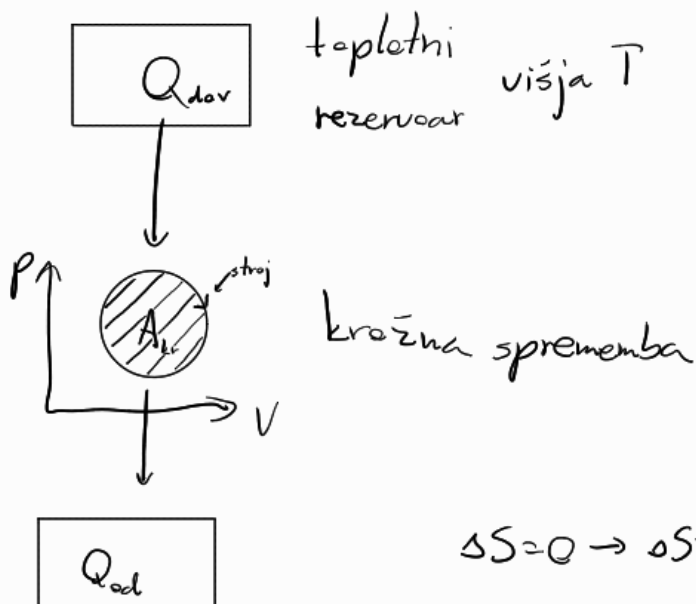




$$\Delta S = 0 \rightarrow \oint \frac{dQ_{dov}}{T'} - \oint \frac{|dQ_{adv}|}{T} = 0$$

$$\Rightarrow T' > T$$

Carnot



$$A_{kr} = \oint p dV$$

↓

$$\Delta W_u = 0 = A + Q_{dov} - |Q_{adv}|$$

$$|A_{kr}| = Q_{dov} - |Q_{adv}|$$

$$Q_{dov} = \frac{m}{M} R T_2 \ln \frac{V_2}{V_1}$$

$$Q_{adv} = \frac{m}{M} R T_1 \ln \frac{V_4}{V_3}$$

$$\left. \begin{array}{l} T_2 V_2^{\gamma-1} = T_1 V_3^{\gamma-1} \\ T_2 V_1^{\gamma-1} = T_1 V_4^{\gamma-1} \end{array} \right\} \frac{V_2}{V_1} = \frac{V_4}{V_3}$$

$$\Rightarrow \frac{Q_{dov}}{T_2} = \frac{|Q_{adv}|}{T_1}$$

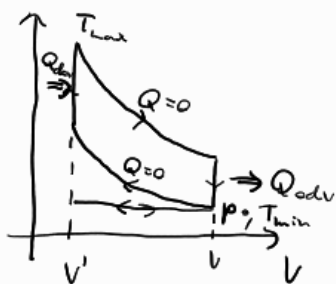
Izkoristek

$$\eta = \frac{|A_{kr}|}{Q_{dov}} = 1 - \frac{|Q_{adv}|}{Q_{dov}} = 1 - \frac{T_1}{T_2}$$

Carnotov topl. stroj $\rightarrow$ idealni toplotni stroj

(ni možno narediti topl. str. pri dani temp. razl. z boljšim izkoristkom)

Bencinski motor

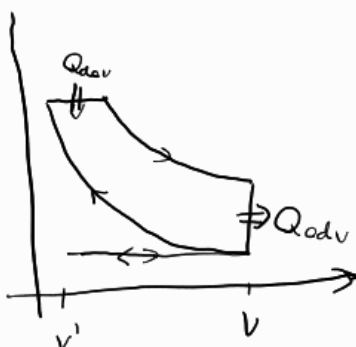


η - polovica Carnotovega izk.

$$\eta \approx 0,35$$

$$\kappa = \frac{V}{V'}$$

Dizelski motor

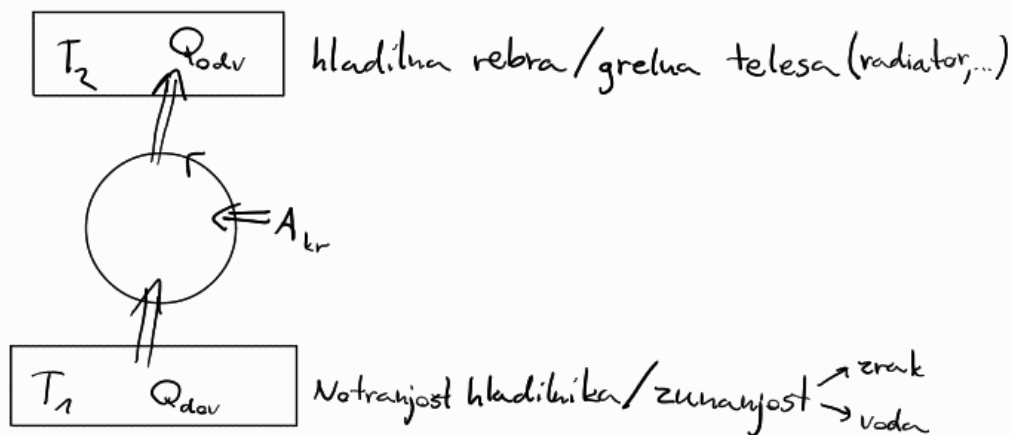


čeprav je $\frac{T_2}{T_1}$ pri bencinu večje, je izkoristek boljši pri dizelskem motorju zaradi večjega κ

$$P \propto \phi_{m,O_2}$$

$$\frac{P}{P_0} \cdot V_{mot} \cdot \omega$$

Hladilnik in toplotna črpalka



Zmogljivost

$$\eta_{TC} = \frac{|Q_{odv}|}{A_{kr}} = \frac{|Q_{odv}|}{|Q_{odv}| - Q_{dav}} \rightarrow \eta_{TC,p} = \frac{T_2}{T_2 - T_1} > 1$$

Carnotova/idealna

Primer:
standard
za zmogljivost
t.c.:

$$\begin{array}{c} T_1 \text{ (zrak)} \quad T_2 \text{ (voda)} \\ A7W35 \\ \downarrow \\ \eta_{TC} = \frac{300K}{30K} \approx 10 \end{array}$$

$$\eta_H = \frac{Q_{\text{dov}}}{A_{\text{br}}} = \frac{Q_{\text{dov}}}{|Q_{\text{odv}}| - Q_{\text{dov}}} \rightarrow \boxed{\eta_{H,C} = \frac{T_1}{T_2 - T_1} > 1}$$

↑
Carnotova/idealna

Primer: isti podatki kot prej:

$$\eta_{H,C} = \frac{270\text{ K}}{30\text{ K}} = 9$$

Hlajenje tekočega He:

$$\eta_{H,C} = \frac{4\text{ K}}{300\text{ K}} \rightarrow \underline{\text{zelo drago!}}$$